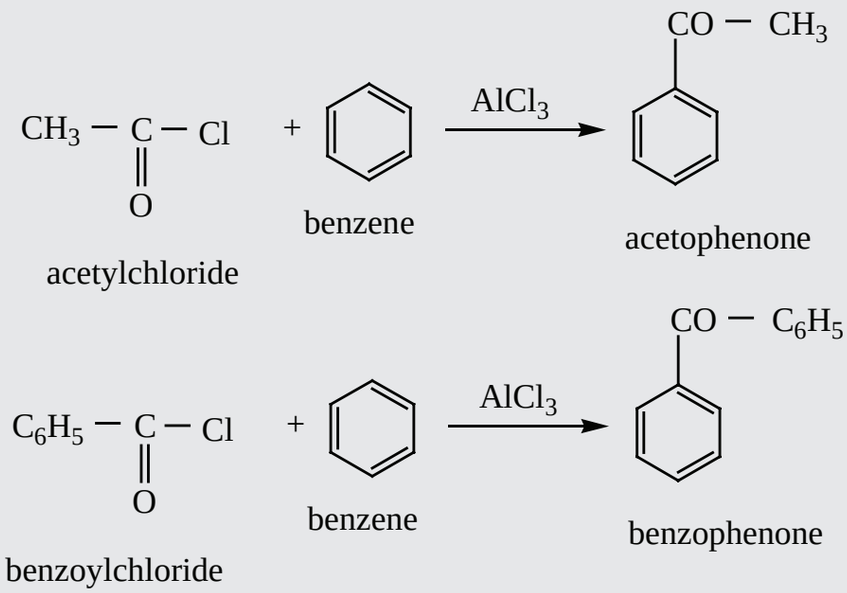
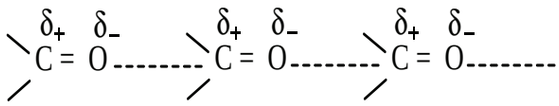
Example



12.4 Physical properties of aldehydes and ketones

1. **Physical State:** Formaldehyde is a gas at room temperature and acetaldehyde is a volatile liquid. All other aldehydes and ketones upto to C_{11} are colourless liquids while the higher ones are solids.
2. **Boiling points**

Aldehydes and ketones have relatively high boiling point as compared to hydrocarbons and ethers of comparable molecular mass. It is due to the weak molecular association in aldehydes and ketones arising out of the dipole-dipole interactions.



These dipole-dipole interactions are weaker than intermolecular H-bonding. The boiling points of aldehydes and ketones are much lower than those of corresponding alcohols and carboxylic acids which possess inter molecular hydrogen bonding.

Compound	Molar mass	Boiling point (K)
$CH_3(CH_2)_3CH_3$ Pentane	**72**	**309**
$CH_3(CH_2)_2CHO$ butanal	**72**	**349**
$CH_3(CH_2)_3OH$ butanol	**74**	**391**

Compound	Molar mass	Boiling point (K)
$CH_3CH_2COCH_3$ butan - 2- one	**72**	**353**
CH_3CH_2COOH Propanoicacid	**74**	**414**

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